Replication and Extension of Outcome Bias in Sports: Evidence from the Indian Premier League (IPL)

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I. INTRODUCTION

This term paper replicates and extends the empirical analysis in *Replication: Do coaches stick with what barely worked? Evidence of outcome bias in sports* by Pascal Flurin Meier, Raphael Flepp, and Egon Franck (2023). The paper revisits and broadens earlier evidence on outcome bias in sports decision-making, asking whether coaches systematically respond differently to narrow wins and narrow losses even when those outcomes may be driven in part by chance. In particular, the authors study whether coaches are more likely to retain a starting lineup after a narrow win than after a narrow loss, despite the fact that such close outcomes may not provide strong evidence that one underlying strategy was truly superior. This setting provides a clean and intuitive context for testing outcome bias: when outcomes are close, decision-makers may over-attribute success or failure to the quality of the prior decision rather than to randomness.

The original paper is important because it speaks directly to a central question in behavioral economics and economic psychology: do decision-makers revise strategies rationally, or do they place excessive weight on realized outcomes? In standard models of rational updating, narrow wins and narrow losses should not generate dramatically different responses if the underlying performances were very similar. However, if outcome bias is present, decision-makers may treat a narrow win as validation and a narrow loss as evidence that change is necessary. Meier, Flepp, and Franck (2023) show that this pattern appears in professional sports, where coaches are less likely to change lineups after narrow wins than after narrow losses, providing evidence that coaching decisions may be shaped by outcome bias rather than purely by underlying performance.

In the first part of this term paper, I replicate the original paper using the authors' replication package and confirm the central empirical finding that narrow wins tend to be associated with fewer subsequent lineup changes than narrow losses. This replication is valuable because it verifies that the original results are reproducible and provides a foundation for a related extension in a new sports context. The ability to reproduce the original results also increases confidence that the underlying empirical design and coding choices are transparent and robust.

In the second part of the paper, I extend the analysis to the Indian Premier League (IPL), using match-level cricket data to examine whether teams are more likely to change their opening batting lineup after narrowly losing rather than narrowly winning. This is a natural extension because, although cricket differs from basketball and football, the choice of opening batsmen is a highly

salient strategic decision that can meaningfully affect match outcomes. In T20 cricket, the opening pair shapes the scoring pace in the powerplay, influences early risk-taking, and can determine how the rest of the batting order is deployed. As a result, changes to the opening lineup provide a useful analogue to the lineup adjustment decisions studied in the original paper.

To implement the extension, I construct an indicator for whether a team changes its opening batting pair relative to the previous match and relate this decision to the outcome of the previous game. Following the logic of the original paper, I focus especially on close matches defined by small run margins (within 5, 10, and 15 runs). I estimate linear probability models that control for match outcome, score difference, and team fixed effects, and I also include a robustness check with season fixed effects. The extension yields a pattern consistent with the original paper's intuition: teams appear directionally more likely to change their opening pair after losses than after wins, especially in close games. However, unlike the original paper, these effects are not statistically significant in the IPL sample. Overall, the extension provides suggestive but not conclusive evidence that outcome bias may also influence strategic lineup decisions in cricket.

II. OVERVIEW OF ORIGINAL PAPER AND EXTENSION

The original paper I replicate is *Replication: Do coaches stick with what barely worked? Evidence of outcome bias in sports* by Meier, Flepp, and Franck (2023). This study revisits the earlier result in Lefgren, Platt, and Price (2015), who document evidence of outcome bias in coaching decisions. Outcome bias refers to the tendency to evaluate a prior decision based too heavily on its realized outcome rather than on the quality of the decision ex ante. In the sports context, this means that coaches may interpret a narrow win as evidence that their previous lineup choice was correct, while interpreting a narrow loss as evidence that change is needed, even when the two outcomes are very similar and may differ partly because of luck.

The central empirical idea in Meier et al. (2023) is straightforward but powerful. When a team narrowly wins or narrowly loses, the realized outcome is often only weakly informative about whether the coach's prior strategic choice was actually better or worse. If coaches respond rationally only to meaningful information, narrow wins and narrow losses should not lead to sharply different rates of lineup revision. However, if coaches are affected by outcome bias, then they may be less likely to change the lineup after a narrow win and more likely to change it after a narrow loss. The paper therefore tests whether lineup changes exhibit a discontinuous jump at the win-loss threshold in close games.

To study this question, Meier et al. (2023) use a regression discontinuity design (RDD), building directly on the identification strategy in Lefgren et al. (2015). The running variable is the prior

game's point margin, centered at zero, with the cutoff separating narrow wins from narrow losses. The key outcome is whether the coach changes the starting lineup in the subsequent game, as well as the number of player changes in some specifications. The identifying assumption is that teams just above and just below the zero-margin threshold are similar in underlying quality and performance, so that narrow wins and narrow losses are conditionally comparable. Under this design, any discontinuity in subsequent lineup changes at the threshold can be interpreted as evidence that coaches react differently to the realized outcome itself rather than to materially different underlying information.

A major contribution of Meier et al. (2023) is that the paper does not merely reproduce the original NBA result. Instead, it conducts what the authors describe as a broader scientific replication. First, the authors extend the data horizon in the NBA relative to the original study. Second, they test whether the same pattern appears in additional basketball settings, including the WNBA and college basketball. Third, they examine whether the phenomenon generalizes beyond basketball by studying the NFL. Fourth, they assess whether the strength of the effect changes over time. Finally, they implement a more modern regression discontinuity framework using recent methodological advances associated with Calonico, Cattaneo, Farrell, and Titiunik, including bias-corrected inference and improved bandwidth selection. This is an important improvement because it strengthens the credibility of the identification strategy and places the replication in line with current best practice for RD estimation.

The main findings of the paper strongly support the original interpretation. Meier et al. (2023) successfully replicate the NBA result from Lefgren et al. (2015): coaches are more likely to change their starting lineup after narrow losses than after narrow wins. The paper then shows that similar patterns appear in other basketball contexts and in the NFL, suggesting that the phenomenon is not unique to one league or one sport. Overall, the evidence indicates that outcome bias in coaching decisions is robust across multiple settings, although the paper also notes that the strength of the effect appears to have weakened over time in some samples. In this sense, the paper both confirms the original result and broadens its external validity, while also contributing to the growing literature that uses sports settings to study behavioral biases in high-stakes field environments.

In the first stage of my term paper, I use the authors' replication package to reproduce the main empirical results from Meier et al. (2023). This replication exercise is important for two reasons. First, it confirms that the published results are computationally reproducible using the provided code and data. Second, it gives me a clear empirical benchmark for evaluating my own extension. By reproducing the main finding that coaches change lineups more often after narrow losses than after narrow wins, I establish that the outcome-bias mechanism documented in the original paper is present in the original setting before asking whether a similar pattern can be observed in professional cricket.

This replication naturally motivates my extension to the Indian Premier League (IPL). While the original paper studies lineup revision in basketball and football, the same behavioral logic may plausibly apply in cricket. In T20 cricket, the opening batting pair is a central strategic choice that shapes the powerplay, early scoring rate, and the deployment of the rest of the batting order. As a result, a coach or team management group may be tempted to interpret a narrow win as confirmation that the existing opening pair "worked," while viewing a narrow loss as a signal that a change is necessary. The next section explains how I adapt the original paper's logic to IPL data and test whether teams are more likely to change their opening batting lineup after narrowly losing rather than narrowly winning.

III. IPL DATA AND VARIABLE CONSTRUCTION

For the extension, I constructed a team-match level dataset using IPL match data that includes the identity of each team, the date of the match, the match outcome, the run margin, and the names of the two opening batsmen. Each observation corresponds to a team in a given match. This structure allows lineup choices to be tracked over time within teams.

The key dependent variable is lineup_change, which is an indicator equal to 1 if a team changed its opening batting pair relative to its previous match, and 0 otherwise. To construct this variable, I sorted observations by team and match date, then compared the current match's opener1 and opener2 to the previous match's opening pair for the same team. If either opener differed from the prior match, lineup_change was coded as 1. The first observation for each team has no prior match within the sample, so lineup_change is missing for those cases.

The main explanatory variable is won, an indicator for whether the team won its previous match. To capture closeness of the prior result, I also constructed a signed run-margin variable, score_diff, which equals the run margin if the team won and the negative of the run margin if the team lost. Thus, positive values represent wins and negative values represent losses, with values near zero corresponding to close matches.

To focus on narrow outcomes, I then created three close-game indicators:

- close5 = 1 if the absolute value of score_diff is less than or equal to 5
- close10 = 1 if the absolute value of score_diff is less than or equal to 10
- close15 = 1 if the absolute value of score_diff is less than or equal to 15

These alternative bandwidths serve the same purpose as varying the window around the cutoff in a regression discontinuity design: narrower windows provide a cleaner comparison between very

similar wins and losses, while wider windows increase sample size at the cost of including less comparable matches.

In addition, because team and season are stored as string variables in the raw data, I encoded them into numeric identifiers (team_id and season_id) so they could be used in fixed-effects regressions in Stata.

The original paper uses a regression discontinuity design centered on the narrow win-loss threshold. Because my IPL extension is constructed manually from available match-level data rather than from the original authors' full RD framework, I implement a simpler but conceptually similar empirical strategy. Specifically, I compare lineup changes after close wins and close losses within small bandwidths around zero run margin.

My baseline empirical model is a linear probability model of the form:

$$\text{lineup_change}_{it} = \alpha + \beta\,\text{won}_{it-1} + \gamma\,\text{score_diff}_{it-1} + \delta_i + \varepsilon_{it}$$

where:

- $\text{lineup_change}_{it}$ indicates whether team i changes its opening pair in match t,
- won_{it-1} indicates whether team i won its previous match,
- score_diff_{it-1} is the signed run margin in the previous match,
- δ_i represents team fixed effects,
- and the sample is restricted to close games using the close5, close10, or close15 indicators.

The coefficient of interest is β. A negative estimate of β means that teams are less likely to change their opening pair after a win, which is equivalent to saying that teams are more likely to change after a loss. This is the pattern predicted by outcome bias.

Including score_diff helps account for the exact magnitude of the previous result within the close-game window, while team fixed effects absorb persistent differences across franchises in how frequently they rotate their opening pair. For example, some teams may systematically experiment more than others because of coaching style, squad depth, injuries, or strategic philosophy. The team fixed effects therefore allow the estimates to be driven by within-team comparisons over time.

As a robustness check, I also estimate a richer specification that includes season fixed effects in addition to team fixed effects. This helps control for season-specific differences such as rule changes, evolving league conditions, roster churn, or broader changes in strategic norms over time.

The season fixed-effects specification is especially useful because the IPL spans many seasons and includes both changes in team composition and changes in league structure.

Although this empirical strategy is not a full local linear RD with bias-corrected inference like the original paper, it preserves the key identifying intuition: by restricting attention to narrow wins and narrow losses, the analysis compares teams after outcomes that are relatively similar in underlying performance. This makes the extension conceptually aligned with the original outcome-bias framework, while remaining feasible given the available IPL dataset and the scope of the term project.

IV. FINDINGS AND ANALYSIS

I first examine the raw relationship between prior match outcomes and subsequent opening-lineup changes in the full IPL sample. Figure 1 plots the average value of lineup_change separately for teams coming off a loss and teams coming off a win.

Figure1

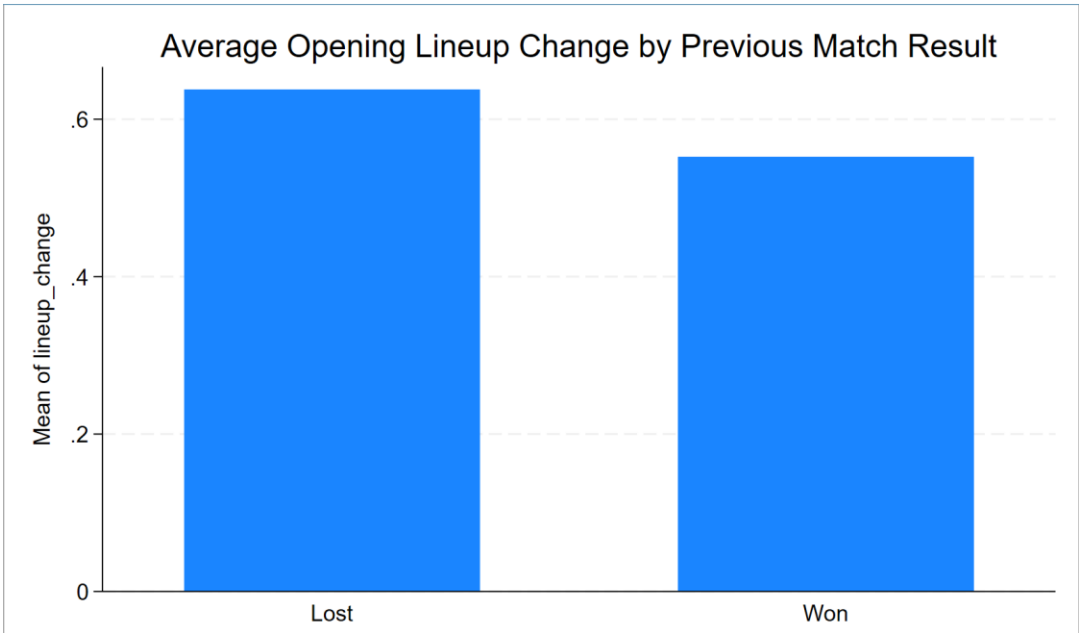

Figure 1 shows a clear descriptive pattern: teams are more likely to change their opening batting pair after a loss than after a win. In the full sample, the average probability of changing the opening pair is approximately 0.64 after a loss and 0.55 after a win. This is a sizable descriptive difference

and is consistent with the basic intuition of the original paper. In other words, even before conditioning on closeness, IPL teams appear to react more aggressively to negative outcomes than to positive ones.

Because the core logic of outcome bias is most compelling in settings where wins and losses are very similar in underlying performance, I then restrict attention to narrow matches. Figure 2 presents the same comparison, but only for matches decided within 5 runs.

Figure 2

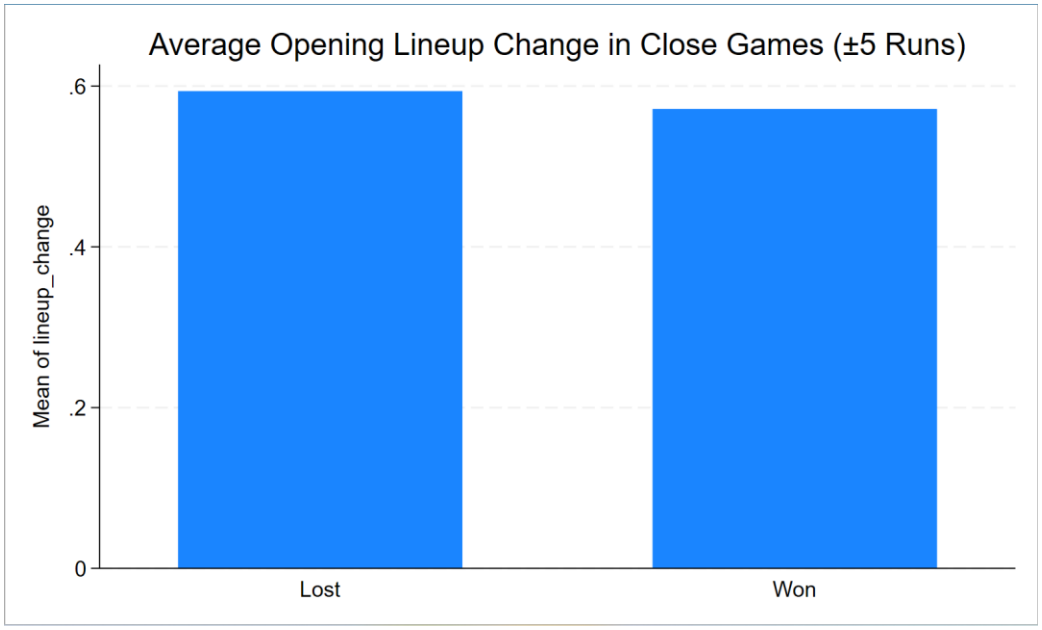

Figure 2 shows that the same directional pattern remains in the narrow-match sample, although the difference is smaller. In matches decided within 5 runs, the average probability of changing the opening pair is approximately 0.59 after a loss and 0.57 after a win. This suggests that even in very close games, teams that narrowly lose may be slightly more likely to revise their opening lineup than teams that narrowly win. However, the gap is modest, which foreshadows the weaker statistical evidence in the regression analysis.

Taken together, the descriptive figures provide an intuitive visual summary of the extension. They show that the raw data are broadly consistent with the behavioral mechanism studied in Meier et al. (2023): negative outcomes are associated with more lineup revision than positive outcomes. At the same time, the narrower close-game comparison suggests that the effect may be smaller once attention is restricted to more comparable wins and losses.

I next estimate linear probability models of the probability of changing the opening pair, restricting the sample to close matches defined by alternative run-margin bandwidths. In each specification,

I regress lineup_change on the prior match outcome (won), the signed run margin (score_diff), and team fixed effects. The coefficient of interest is the coefficient on won. A negative coefficient means that teams are less likely to change the opening pair after a win, or equivalently, more likely to change after a loss.

Table 1 reports the first specification, which restricts the sample to matches decided within 5 runs.

TABLE 1—OPENING LINEUP CHANGES AFTER CLOSE MATCHES IN THE IPL: MAIN EXTENSION RESULTS

	(1) +/- 5 runs	(2) +/- 10 runs	(3) +/- 15 runs	(4) +/- 5 runs + Season FE
Panel A. Linear probability models of opening lineup changes				
Won previous match	-0.207	-0.100	-0.078	-0.234
	(0.222)	(0.134)	(0.106)	(0.221)
Score difference	0.034	0.008	0.006	0.036
	(0.034)	(0.011)	(0.006)	(0.034)
Team fixed effects	Yes	Yes	Yes	Yes
Season fixed effects	No	No	No	Yes
Observations	127	254	391	127
R-squared	0.112	0.051	0.048	0.308

In this specification, the estimated coefficient on won is -0.2065 with a p-value of 0.354 ($N = 127$). The negative sign implies that teams coming off a narrow win are less likely to change their opening pair than teams coming off a narrow loss. Interpreted literally, the estimate suggests that teams may be about 20.7 percentage points less likely to change openers after a win than after a loss within the 5-run bandwidth. This is economically meaningful in magnitude and strongly consistent with the direction predicted by the original paper. However, the estimate is not statistically significant, and the confidence interval is wide. As a result, the evidence from the narrowest bandwidth should be interpreted as suggestive rather than definitive.

Table 2 expands the bandwidth to include matches decided within 10 runs.

TABLE 2—DESCRIPTIVE EVIDENCE ON OPENING LINEUP CHANGES AFTER WINS AND LOSSES

Sample	After loss	After win	Difference (loss – win)
Full sample	0.64	0.55	0.09
Close games (+/- 5 runs)	0.59	0.57	0.02
Close games (+/- 10 runs)	0.59	0.57	0.02
Close games (+/- 15 runs)	0.59	0.57	0.02

This table presents descriptive averages of the probability that an IPL team changes its opening batting pair in the following match, conditional on whether the team lost or won its previous match. The full-sample row uses all matches in the sample, while the close-games row restricts attention to matches decided within 5 runs. The difference is calculated as the average after losses minus the average after wins.

Overall, the extension provides suggestive but not statistically strong evidence that IPL teams are more likely to change their opening batting lineup after a loss than after a win. The descriptive evidence in Table 2 and Figure 1 shows a clearer difference in the full sample, with lineup changes occurring more often after losses than after wins. However, once attention is restricted to close matches, the gap becomes smaller, as shown in Figure 2. This is consistent with the intuition behind the original paper: when outcomes are very close, the final result may reflect more randomness and should not necessarily drive major strategic changes. The regression results in Table 1 reinforce this pattern. Across all specifications, the coefficient on the previous match being a win is negative, implying that teams are less likely to change openers after a win. At the same time, the estimates are not statistically significant at conventional levels. Therefore, while the signs of the coefficients are broadly consistent with outcome bias, the extension should be interpreted as suggestive rather than conclusive evidence in the IPL setting.

V. CONCLUSION

This term paper replicated the main idea of Meier, Flepp, and Franck (2023), who revisit the question of whether coaches “stick with what barely worked” and interpret this behavior as evidence of outcome bias in sports decision-making. Their replication study shows that the original NBA result does not hold in the NBA itself, but that similar patterns appear in other sports settings, suggesting that outcome bias may still affect coaching decisions more broadly. Using the authors’ replication framework as motivation, I extended the analysis to the Indian Premier League (IPL) by examining whether teams are more or less likely to change their opening batting lineup depending on whether they narrowly won or narrowly lost the previous match.

The descriptive results show that IPL teams appear more likely to change their opening pair after a loss than after a win in the full sample. This pattern remains directionally present in the close-game samples, but the difference becomes smaller when attention is restricted to matches decided by small margins. This is important because close matches are the setting in which the logic of outcome bias is most relevant: if outcomes are partly driven by luck, then a narrow win should not necessarily validate a prior decision, and a narrow loss should not automatically imply that a change is needed.

The regression analysis produces a consistent qualitative result. Across the ±5, ±10, and ±15 run specifications, as well as the ±5 specification with season fixed effects, the coefficient on the previous match being a win is always negative. This means that, conditional on close matches and controlling for score difference and team effects, teams are less likely to change their opening lineup after a win than after a loss. However, these estimates are not statistically significant at conventional levels. As a result, the evidence from the IPL does not provide strong causal confirmation of outcome bias, but it does offer suggestive evidence that the direction of behavior is similar to the broader phenomenon discussed in the literature.

Overall, the IPL extension contributes to the paper by showing how the same conceptual question can be examined in a cricket context, where opening batters play an especially visible and strategically important role. Even though the statistical evidence is weaker than in some of the settings studied by Meier et al. (2023), the extension is still informative because it demonstrates that outcome-based strategic adjustment may also be relevant outside the sports originally studied in the paper. A natural next step for future research would be to use richer cricket data, such as batter-specific performance, opposition quality, venue conditions, or match format, to better distinguish rational tactical adjustments from true outcome bias.

VI. REFERENCES

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